

Name - Atish Kadam
Roll - CS25MTECH14003

Question - 6

6.3 Experiments: Architecture Ablation

Using the dataset:

6.3.1 Depth Ablation

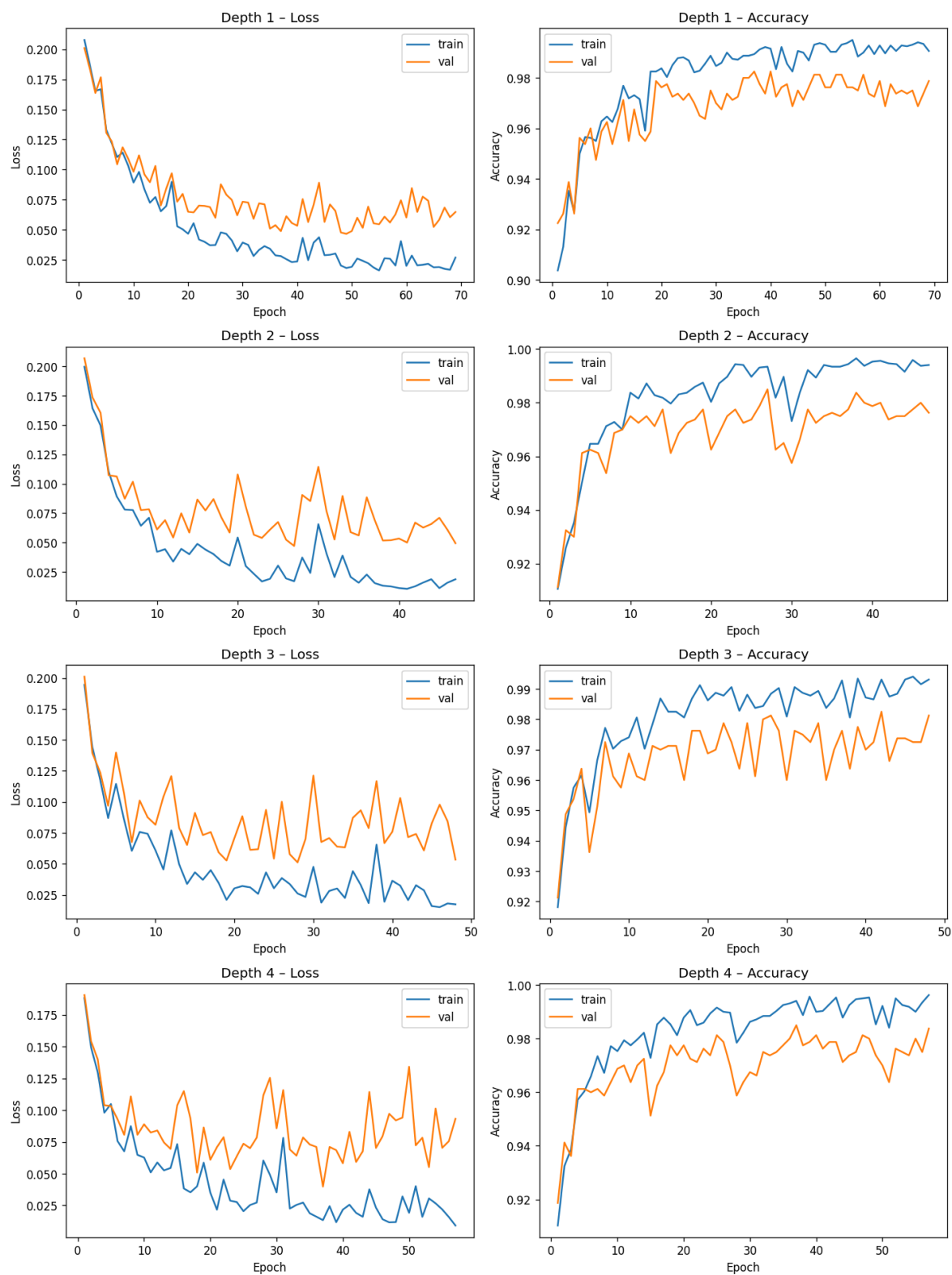


Figure 1: Train/validation loss and accuracy curves for each depth.

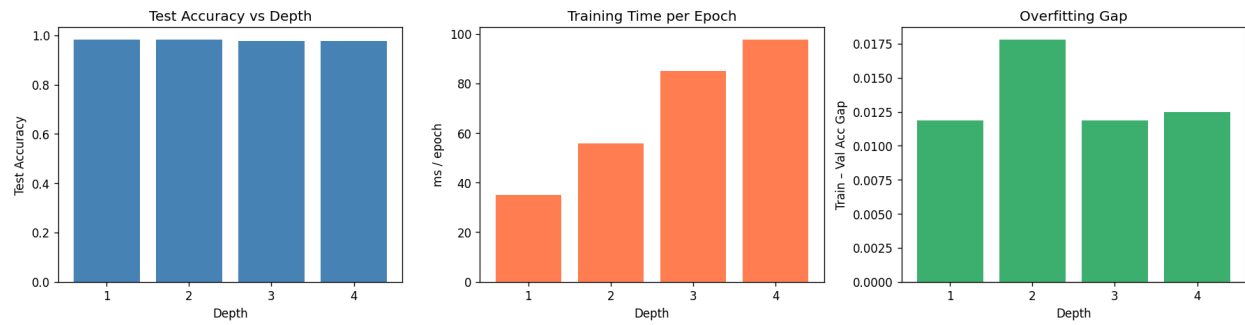


Figure 2: Summary: test accuracy, training time per epoch, and overfitting gap vs. depth.

6.3.2 Width Ablation

1. Fix depth at 2 hidden layers and vary width in the follow ranges:
 - (a) [input dimension, 8, 8, output dimension]
 - (b) [input dimension, 16, 16, output dimension]
 - (c) [input dimension, 32, 32, output dimension]
 - (d) [input dimension, 64, 64, output dimension]
 - (e) [input dimension, 128, 128, output dimension]
 - (f) [input dimension, 256, 256, output dimension]
2. Plot test accuracy vs. number of parameters (log scale)
3. Discuss the width vs. depth performance relationship

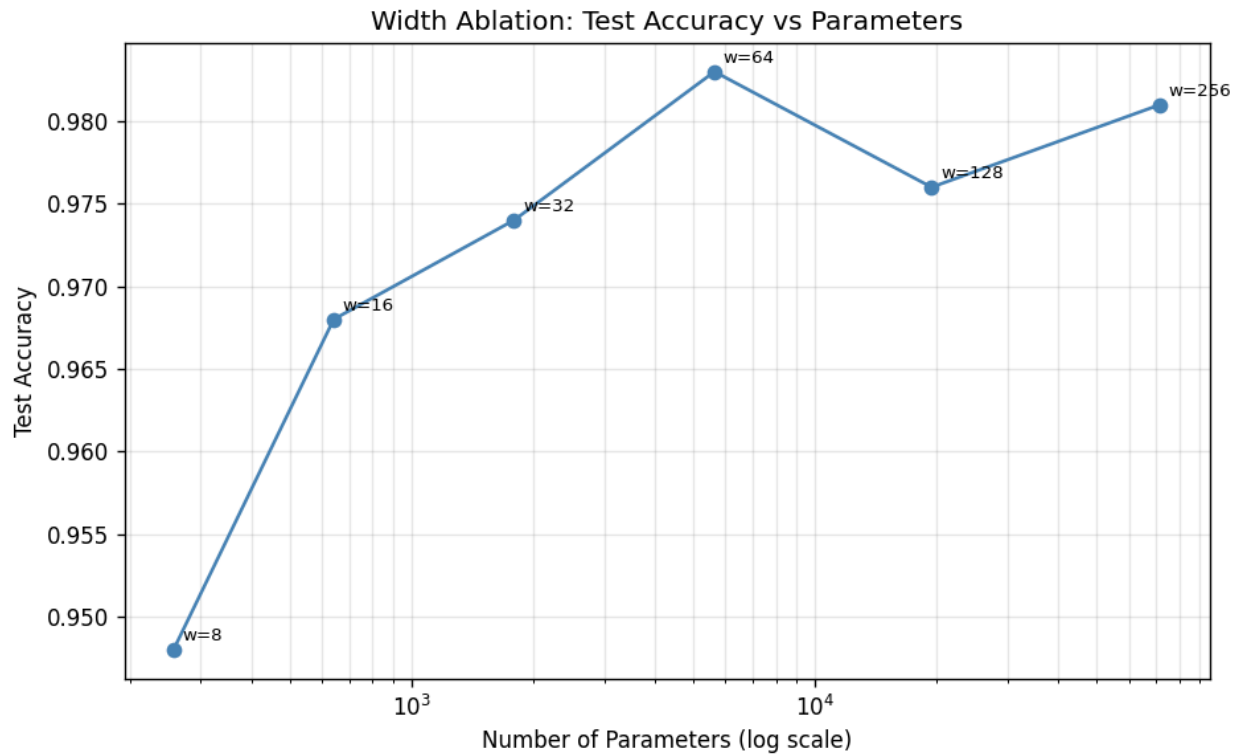


Figure 3: Test accuracy vs. number of parameters (log scale).

6.3.3 Activation Function Comparison

a) Fix architecture at [input, 64, 64, output] and compare:

- Sigmoid in all layers
- Tanh in all layers
- ReLU in hidden layers, sigmoid in output
- Leaky ReLU in hidden layers, sigmoid in output

b) For each configuration:

- Plot loss curves
- Report gradient statistics (mean, std) during training
- Analyze dead neurons (for ReLU variants)

c) Verify vanishing gradients experimentally: measure gradient magnitude at each layer for sigmoid vs. ReLU networks

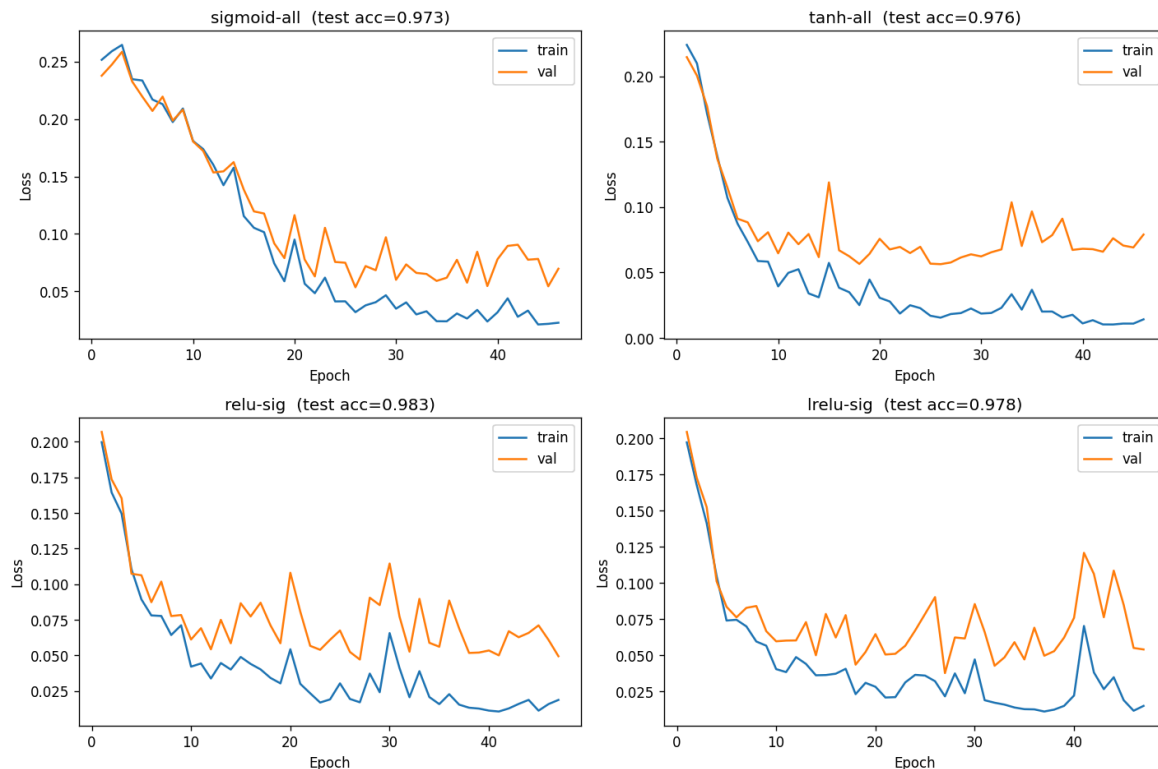


Figure 4: Loss curves for four activation configurations

Gradient Statistics

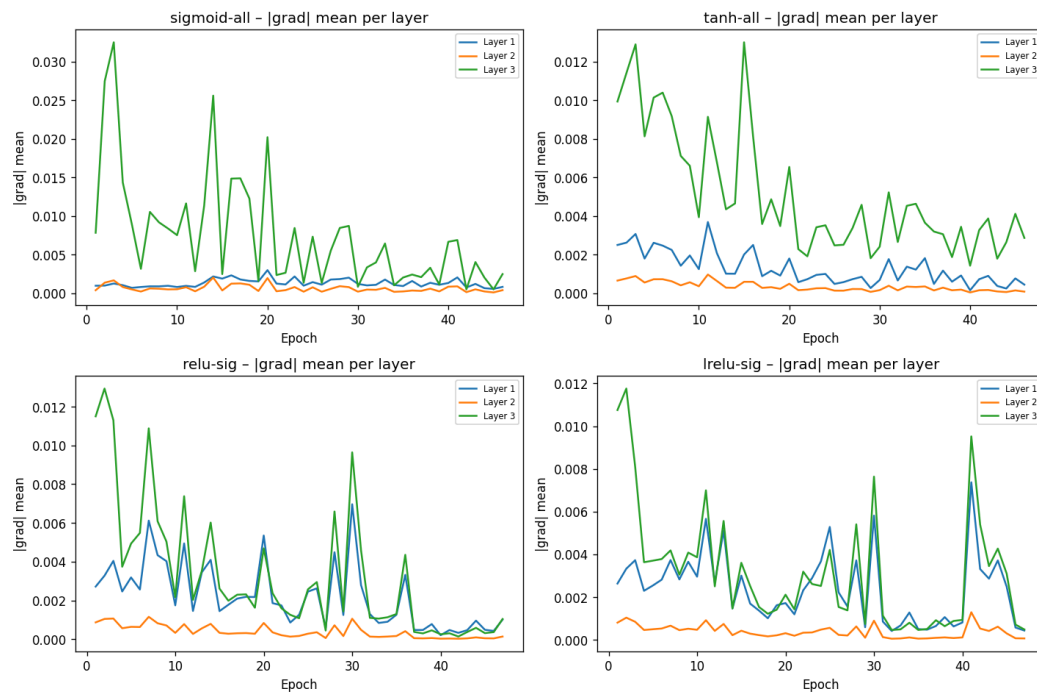


Figure 5: Mean absolute gradient per layer over training.

Vanishing Gradient Verification

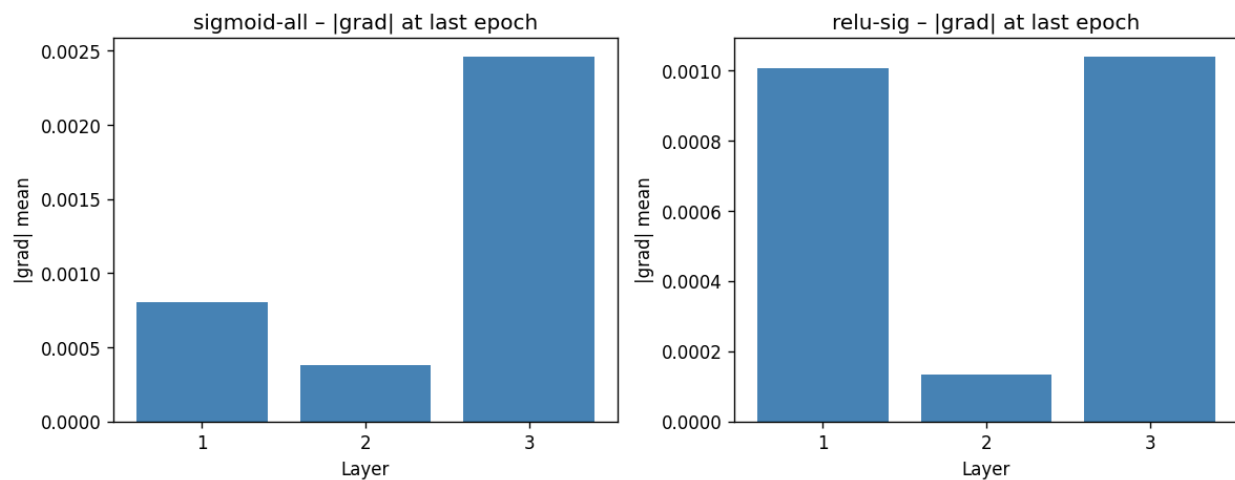


Figure 6: Mean absolute gradient at each layer at the final epoch

Dead Neuron Analysis

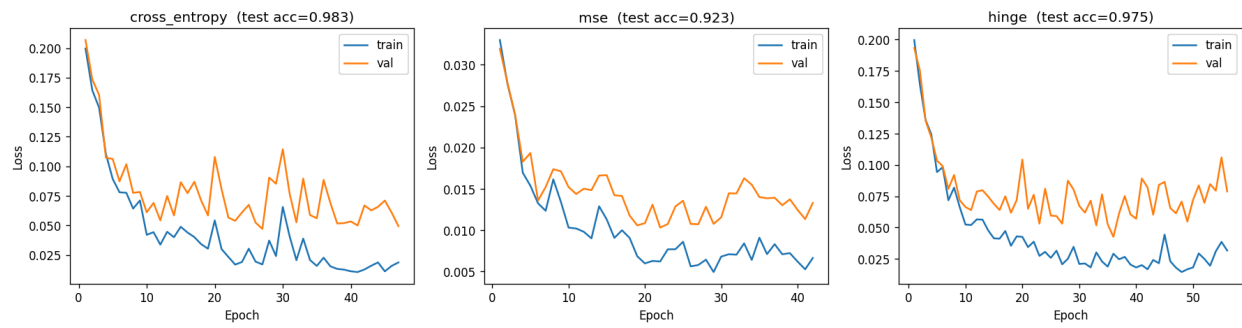


Figure 8: Train/validation loss curves for each loss function.

6.4 Loss Function Analysis

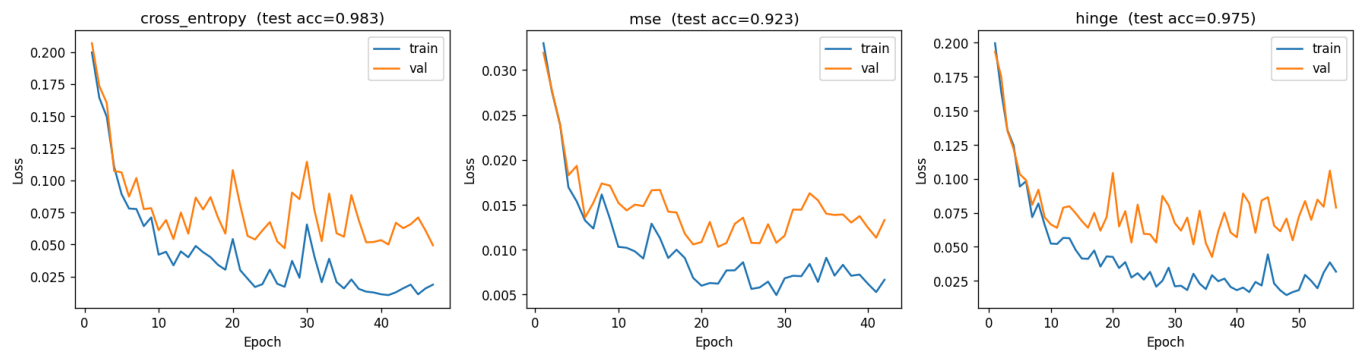


Figure 8: Train/validation loss curves for each loss function.

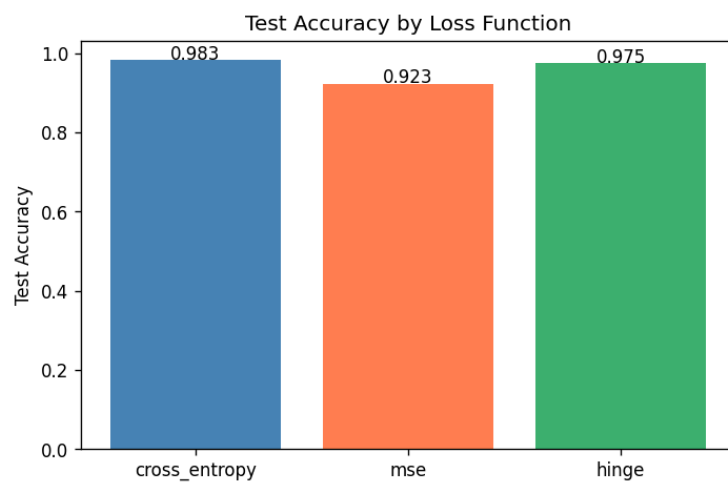


Figure 9: Test accuracy by loss function

6.5 Optimizer Comparison

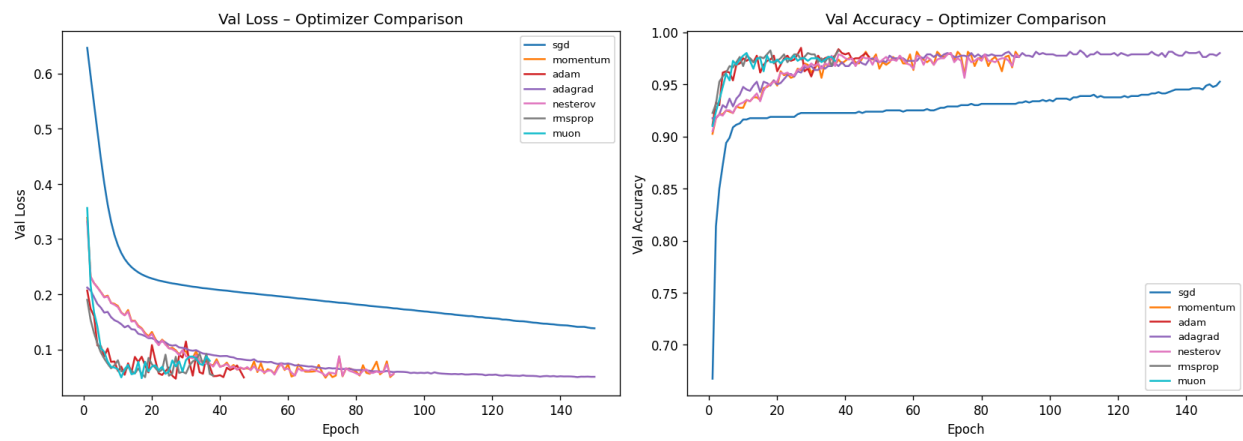


Figure 10: Validation loss and accuracy convergence curves.

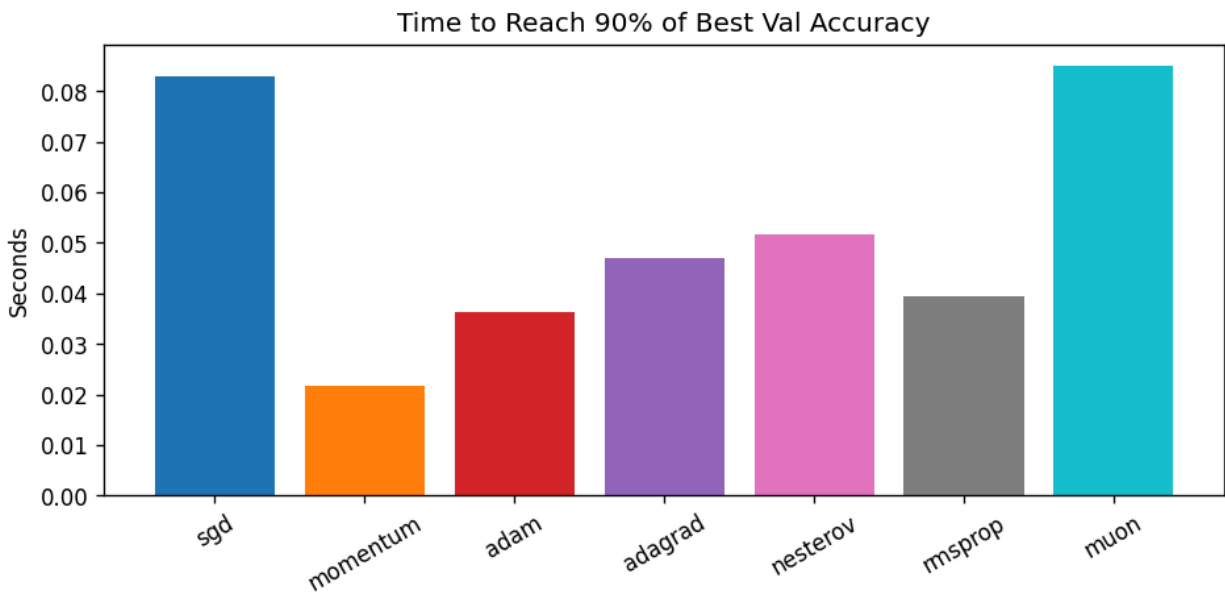


Figure 11: Time (seconds) to reach 90% of best validation accuracy.

Weight Update Magnitudes

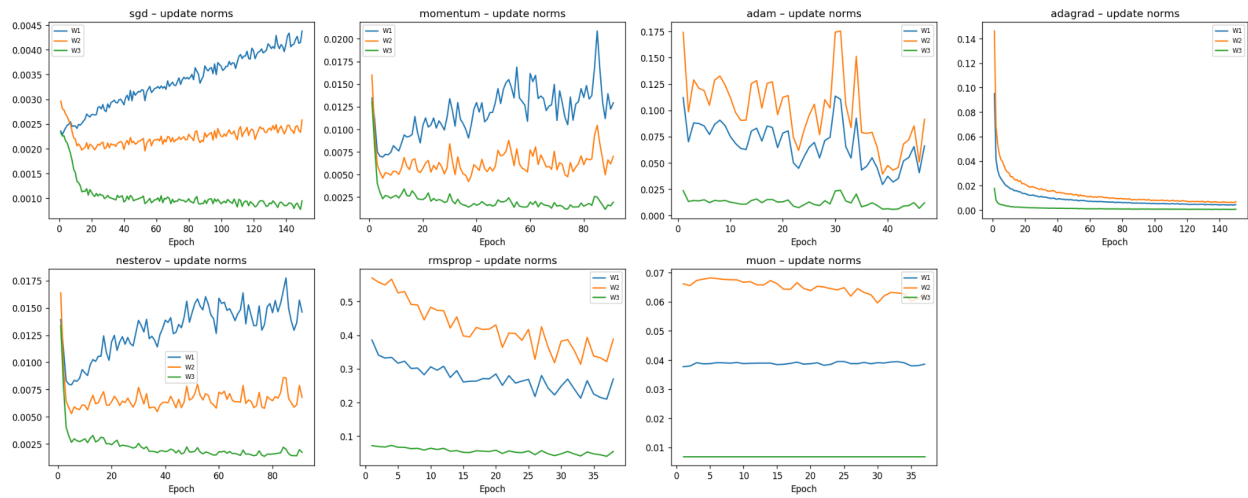


Figure 12: Weight update norm per layer over training for each optimizer

Learning Rate Sensitivity

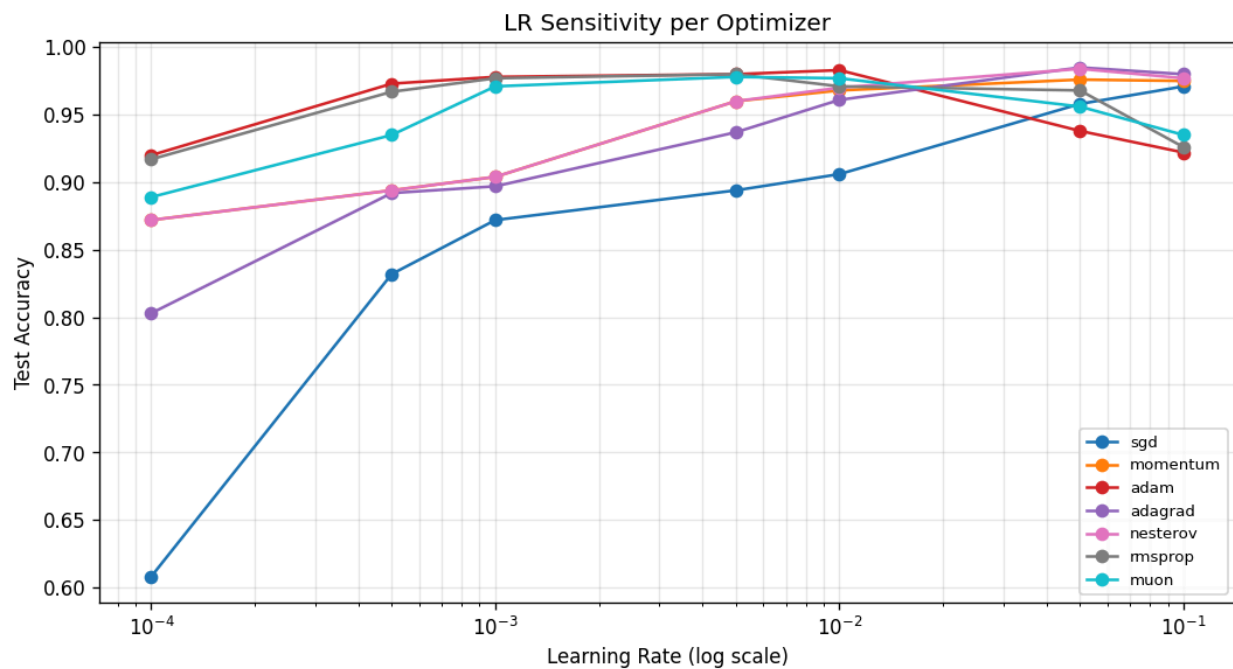


Figure 13: Final test accuracy vs. learning rate for each optimizer.

6.6 Regularization

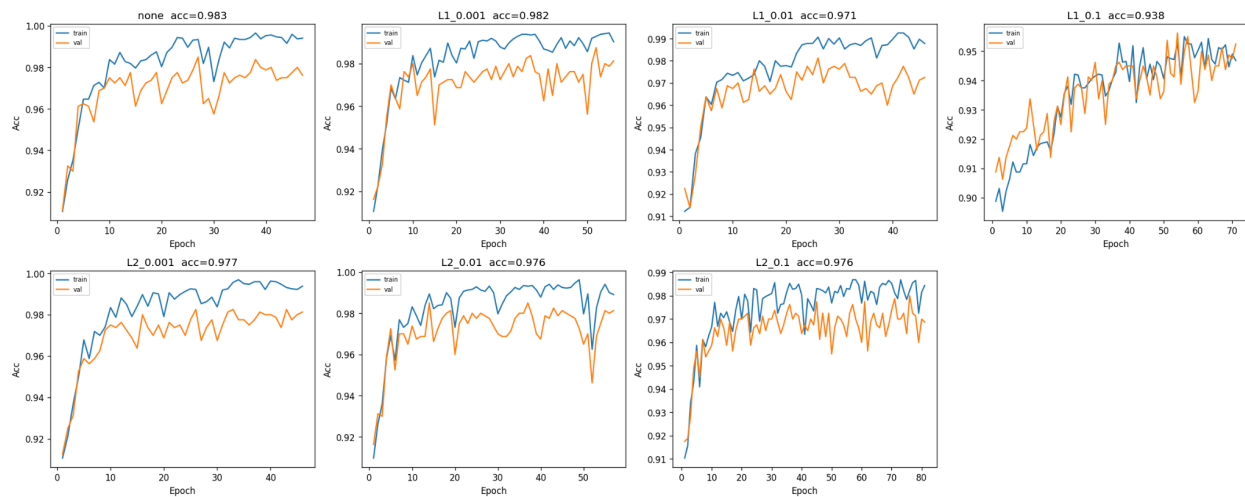


Figure 14: Train vs. validation accuracy for each regularisation setting.

L1 Weight Sparsity

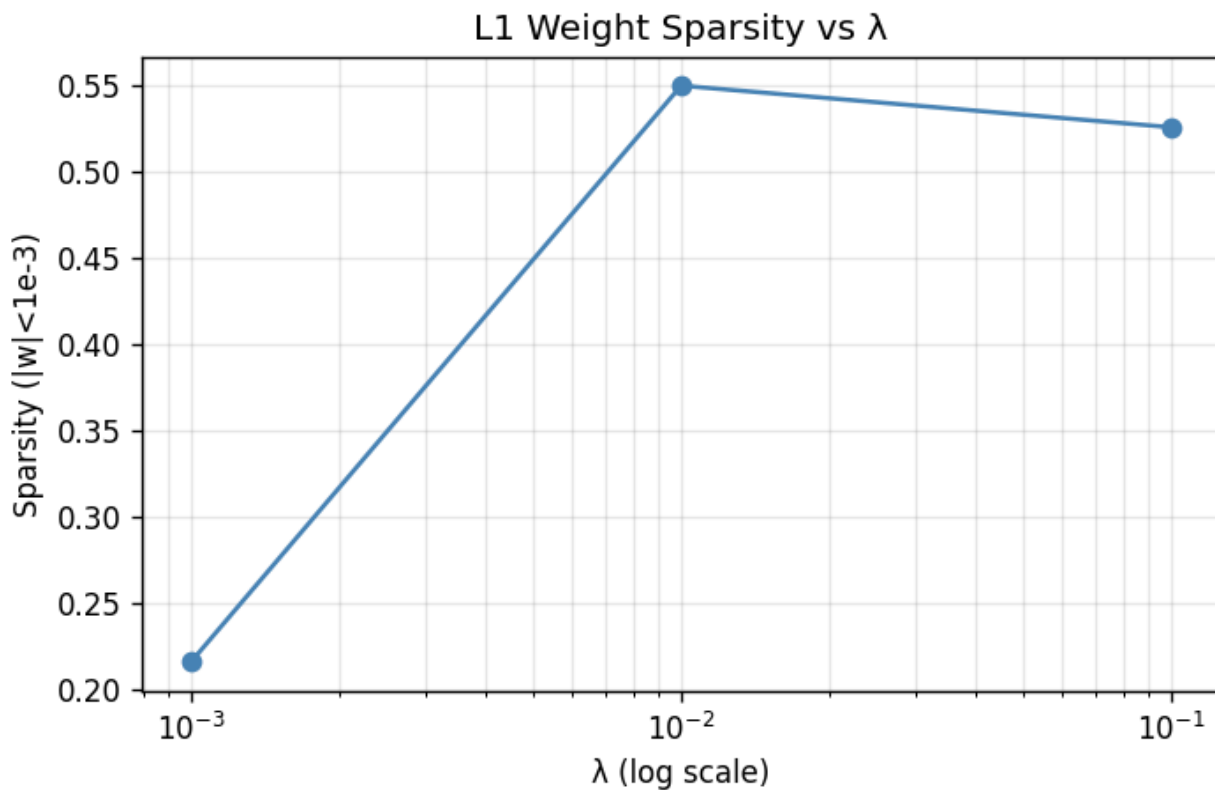


Figure 15: Fraction of weights with $|w| < 10^{-3}$ vs. λ for L1 regularisation

6.7 Success Analysis

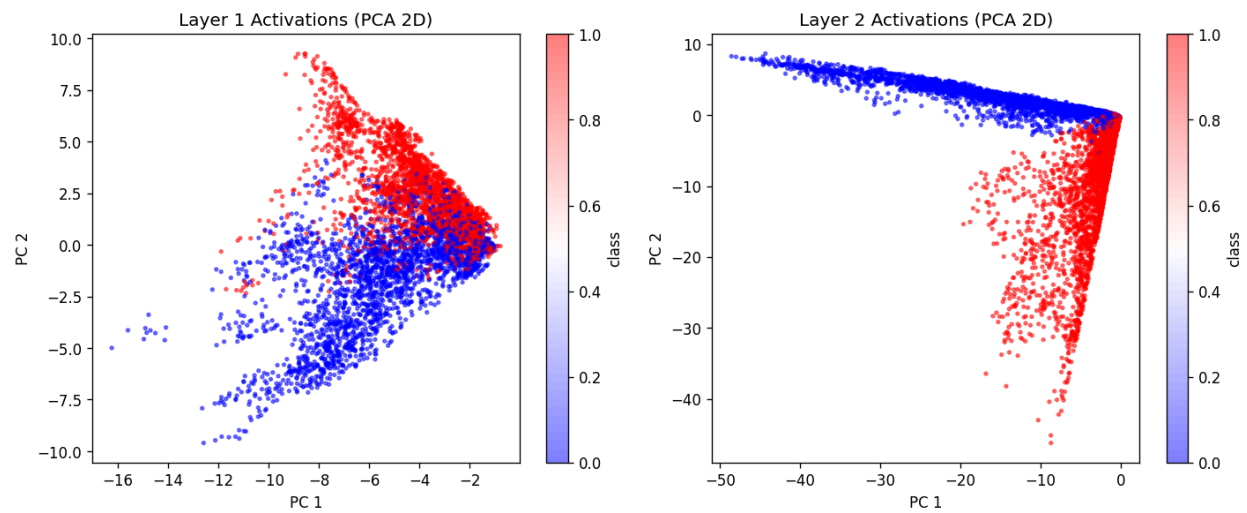


Figure 16: PCA projection of Layer 1 (left) and Layer 2 (right) activations.

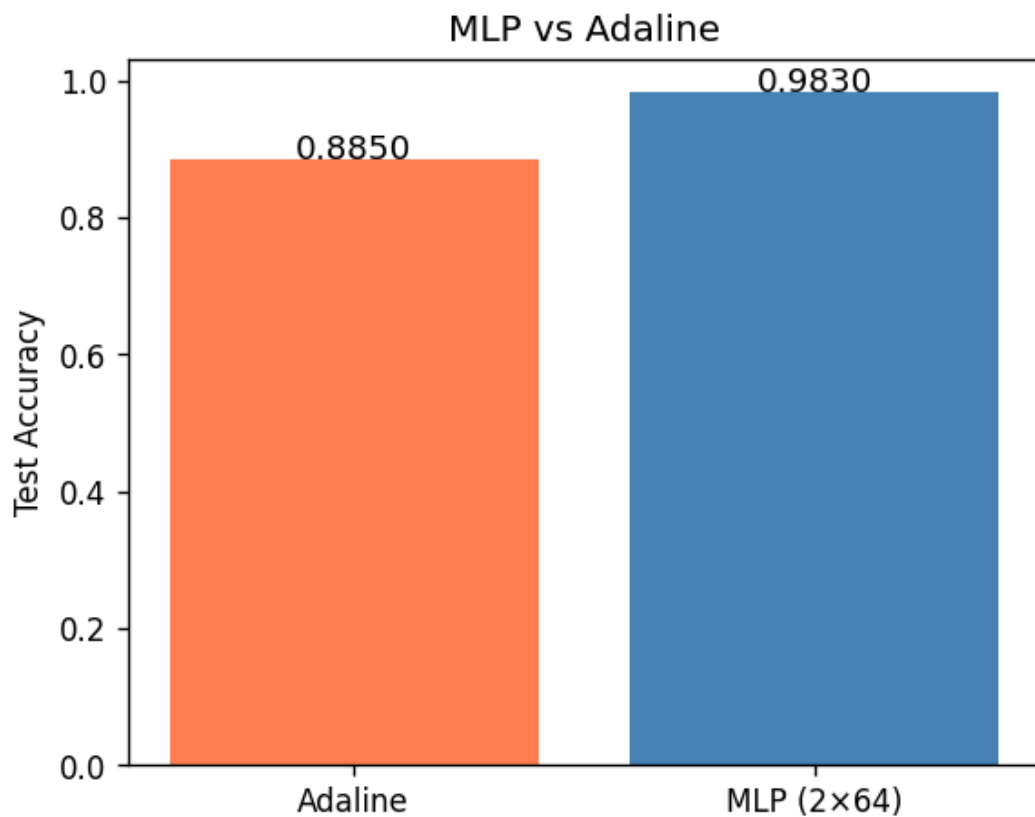


Figure 17: Test accuracy: Adaline vs. MLP with 2 hidden layers.